

MADHAV S KRISHNA

Ernakulam, Kerala | +91 9562621296 | 24me394@mgits.ac.in | linkedin.com/in/madhav-s-krishna

PROFESSIONAL SUMMARY

Mechanical Engineering undergraduate at MITS, with a focused interest in material behaviour, defect analysis, and manufacturing processes. Comfortable leading or contributing within multidisciplinary teams, and driven by an interest in understanding how processing decisions translate into real material and mechanical outcomes.

EDUCATION

B.Tech in Mechanical Engineering (Honours) 2024 – 2028
Muthoot Institute of Technology and Science (KTU), Ernakulam, Kerala

- CGPA: 9.39 / 10 (up to Semester 3)

Higher Secondary & Secondary Education (CBSE) 2017 – 2024
Sree Narayana Public School, Poothotta, Ernakulam

- Class 10: 97% | Class 12: 91.8%

TECHNICAL SKILLS

Manufacturing Quality	&	Casting process analysis, defect identification, root cause analysis, gating system design, solidification principles, process–structure–property relationships
Design & CAD		AutoCAD, Fusion 360, CATIA V5
Instrumentation		Sensor selection and interfacing (current, temperature, vibration), embedded data acquisition, condition monitoring

PROJECTS

Casting System Design and Defect Study using ANSYS 2024 – 2025
Manufacturing Processes — PBL, Semester 3, MITS

Analysed gating system configurations to identify root causes of casting defects including porosity, shrinkage, and misruns. Correlated processing parameters — pouring temperature, gating ratio, and solidification rate — with defect formation outcomes to develop a working understanding of process–structure–property relationships in metal casting. Evaluated design modifications aimed at improving material yield and reducing defect occurrence.

Structural Analysis of Pure and Metal-Doped Crystalline Materials 2025 – Present
Ongoing Capstone Project, MITS

Synthesising crystalline materials via the Sankaranarayanan–Ramasamy (SR) method with controlled metal-ion doping to investigate the influence of dopant concentration on crystal growth behaviour and defect formation. Correlating processing conditions with structural defects to assess their impact on mechanical and functional performance, with systematic experimental documentation maintained throughout.

e-MotorSense — Induction Motor Condition Monitoring System 2025 – 2026
Multidisciplinary Team Project, MITS

Developed a multi-sensor condition monitoring system for induction motors integrating current, temperature, and vibration sensors for real-time anomaly detection and early fault identification. Contributed to sensor selection, placement optimisation, and data interfacing to build a performance monitoring framework applicable to manufacturing equipment reliability.

Autonomous Drone Base Station — ISRO IRoC-U Competition 2026 2025 – Present
Design In-Charge, CoDE Club (Centre for Drone Excellence), MITS

Leading mechanical design of a field-deployable autonomous drone base station, developing architecture

for structural support, secure docking, alignment mechanisms, and energy transfer systems. Currently cleared the Preliminary and Qualifier Rounds.

SAE E-Baja Competition 2026 — Fabrication Junior Engineer

2025 – 2026

All India Rank 32 Overall

Supported vehicle fabrication and mechanical subsystem integration within a structured team environment. Contributed to structural component preparation and fit-up, gaining practical exposure to tolerance management, assembly sequencing, and design-for-manufacturing constraints.

ACHIEVEMENTS

- **IEFK 2026 Finalist** — e-MotorSense selected among Top ~30 of 170+ teams at the International Energy Festival of Kerala, advancing through abstract screening, regional selection, and state-level finals.
- **ISRO IRoC-U Competition 2026** — Qualified Prelims, Qualifiers and currently in Elimination round.
- **Smart India Hackathon 2025** — Selected at college level and submitted the Optimised Onion Storage System project at the national level of SIH 2025.
- **SAE E-Baja 2026** — Secured All India Rank 32 Overall as part of the MITS team.
- **MITS Excellence Award** — Awarded for exceptional excellence at Muthoot Institute of Technology and Science.
- **Department Topper** — Best Academic Performer in Dept. of Mechanical Engineering, MITS.

CERTIFICATIONS

- NPTEL — Product Design and Development: 83% | Design Practice: 79% | Basics of Materials Engineering: 55% (Domain Certifications under Product Design)
- Cambridge Linguaskill Business Test: CEFR Level B2

LEADERSHIP AND RESPONSIBILITIES

- Community Lead — IEDC MITS (Innovation and Entrepreneurship Development Cell)
- Campus Ambassador — Air India Limited
- Media Lead — International Cultural Relations Club (ICRC), MITS
- Public Relations Coordinator — Mahatma Mathrubhumi Study Circle, Thiruvakulam (NGO)

RELEVANT COURSEWORK

Solid Mechanics | Strength of Materials | Manufacturing Processes | Materials Science | Engineering Mechanics | Engineering Mathematics | Thermodynamics